Stock Change Prediction Utilizing Social Media Pools

Michael Petrillo

Colorado State University – Global Campus

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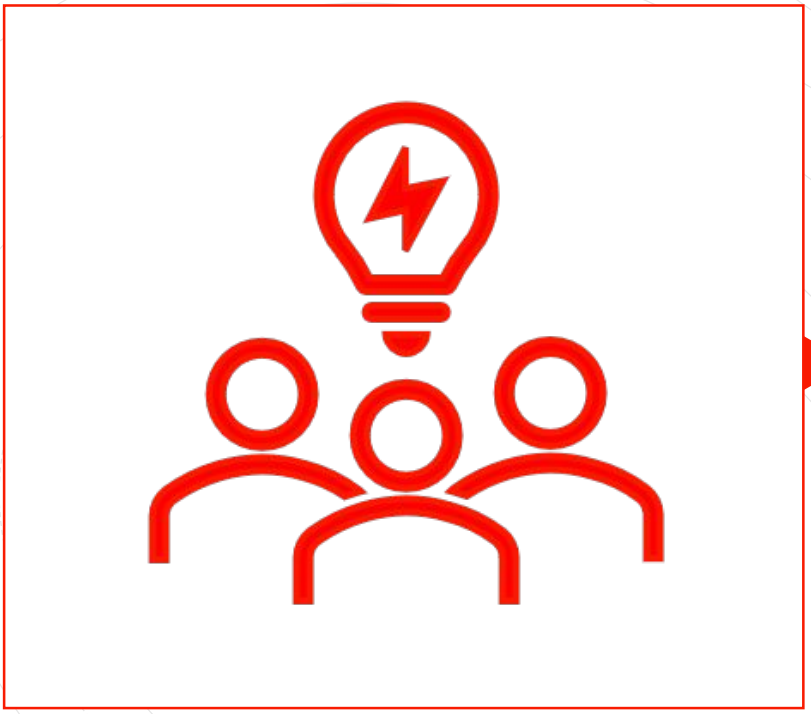


Online interaction has transformed the world's communications over the last decade (Grange, 2018).

Goal

+ User Generated Content (UGC)

+ New York Stock Exchange (NYSE)

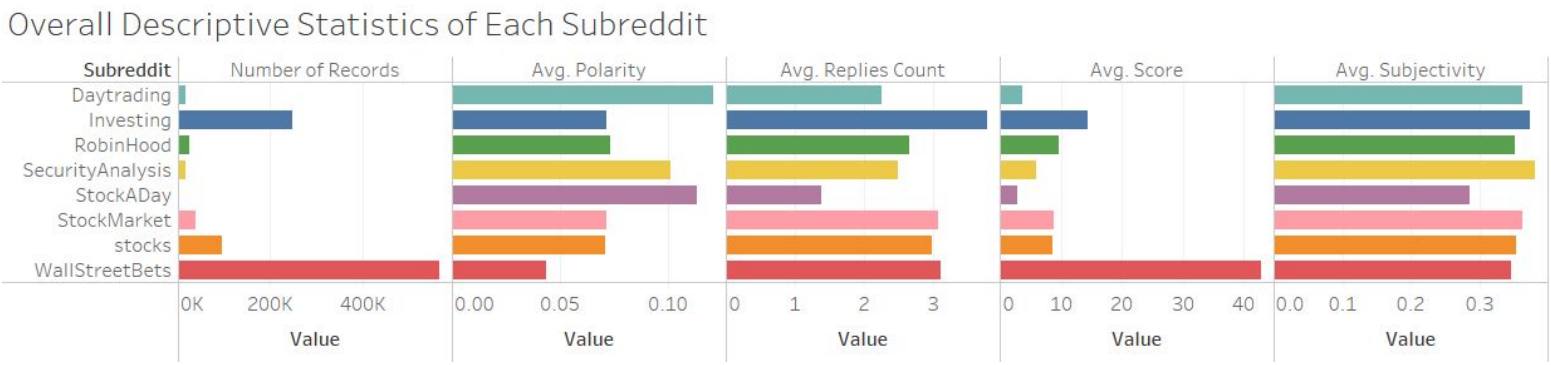
+ Support Vector Machine (SVM)

Accurate Prediction

RESEARCH HYPOTHESIS

Does the UGC increase the accuracy of the SVM model?

- Control – Data only using NYSE
- Experiment – Data merging NYSE and UGC from each subreddit



Subreddits Dataset

NYSE Dataset Metrics

Return Series by hour dataset based on the stock dataset.

- $Delta_{future} = (Close_{future} - Open_{current})/Open_{current}$
- $Delta_{Current} = (Close_{Current} - Open_{previous})/Open_{previous}$
- $Delta_{previous} = (Close_{previous} - Open_{previous})/Open_{previous}$
- $Volume_Delta_{current} = (Volume_{current} - Volume_{previous})/Volume_{previous}$

Data Merge and Accuracy Scoring

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- Data Merge pulls in the Praw and Yfiance data into MySQL then uses Python to create the test and train set before the SVM model is ran.

- Score is calculated by –

$$\text{accuracy}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} 1(\hat{y}_i = y_i)$$

Research Limitations

TIMING

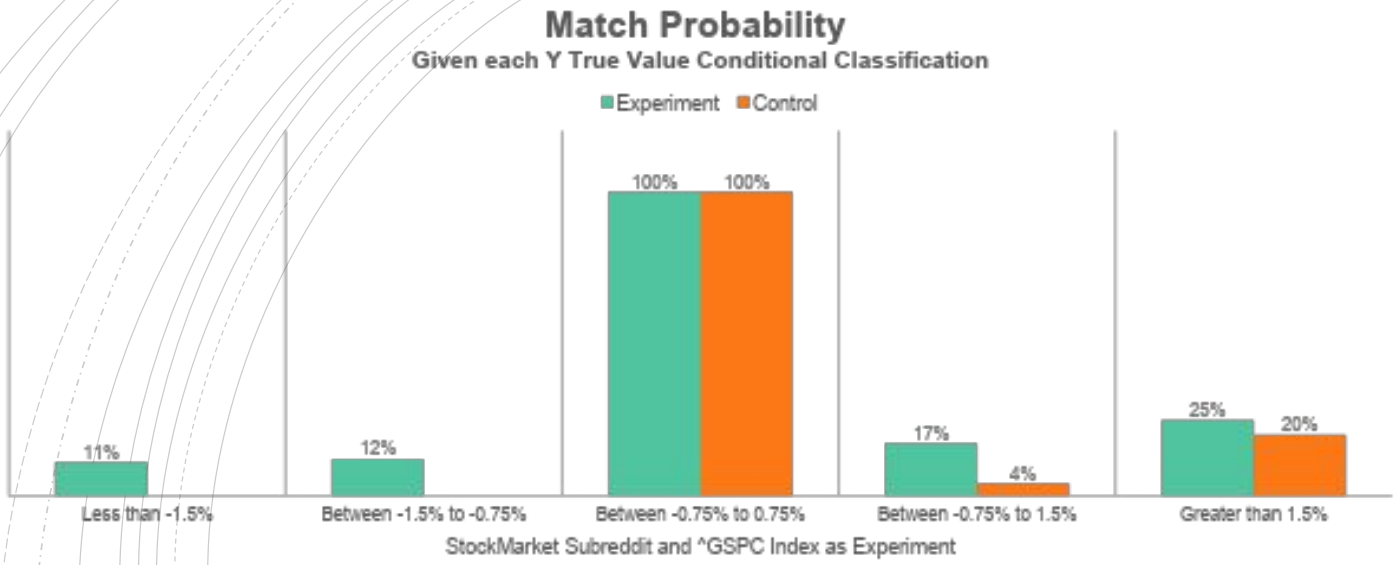
DATA SOURCE

CLASSIFICATIONS

Subreddit And Stock/Index Compared to SVM Skicit Accuracy Score

	Symbol										
Subreddit	^GSPC	WMT	AAPL	AMZN	NFLX	DIS	NVDA	BA	^VIX	TLSA	Grand Total
StockMarket	0.9684	0.9579	0.9579	0.8211	0.8000	0.8316	0.8000	0.6211	0.5158	0.4571	0.7731
Investing	0.9492	0.9576	0.9407	0.8475	0.8305	0.7881	0.7712	0.6017	0.5169	0.4474	0.7651
stocks	0.9684	0.9474	0.9474	0.8211	0.8105	0.8105	0.8000	0.5895	0.4947	0.4521	0.7642
No Subreddit Data	0.9492	0.9492	0.9407	0.8475	0.8305	0.7966	0.7627	0.6102	0.5169	0.4286	0.7632
ALL	0.9492	0.9492	0.9407	0.8390	0.8305	0.7966	0.7797	0.5932	0.5169	0.4286	0.7623
WallStreetBets	0.9407	0.9492	0.9322	0.8305	0.8220	0.7881	0.7627	0.5847	0.5085	0.4286	0.7547
Grand Total	0.9542	0.9517	0.9433	0.8344	0.8207	0.8019	0.7794	0.6001	0.5116	0.4404	0.7638

Findings



Conclusions

Experiment Vs Control Delta of Subreddit and Stock/Index

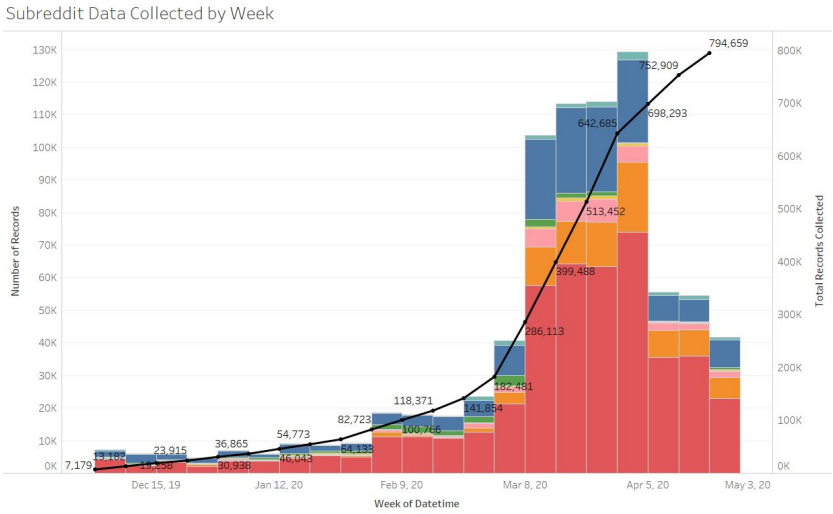
Score based on SVM Accuracy

	Symbol									
Subreddit	^GSPC	WMT	AAPL	AMZN	NFLX	DIS	NVDA	BA	^VIX	TLSA
StockMarket	0.019	0.009	0.017	-0.026	-0.031	0.035	0.037	0.011	-0.001	0.029
stocks	0.019	-0.002	0.007	-0.026	-0.020	0.014	0.037	-0.021	-0.022	0.023
Investing	0.000	0.008	0.000	0.000	0.000	-0.008	0.009	-0.009	0.000	0.019
No Subreddit Data	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ALL	0.000	0.000	0.000	-0.009	0.000	0.000	0.017	-0.017	0.000	0.000
WallStreetBets	-0.009	0.000	-0.008	-0.017	-0.008	-0.008	0.000	-0.025	-0.008	0.000
Grand Total	0.005	0.003	0.003	-0.013	-0.010	0.005	0.017	-0.010	-0.005	0.012

Reference Slide - Overall (Results – Control) as Findings

Field Name	Data Type	Description	Example
comment	string	The text from a reddit comment.	Buying Spy Puts.
score	bigint	The reddit score based on users.	4
author	string	comment author.	somedude_2020
created	datetime	date and time created.	43919.67014
replies_count	bigint	Number of nestled responses.	10
subreddit	string	where did the comment come from in Reddit.	Investing
type	string	Is the data from a comment, title or response.	comment
polarity	double	Text Bolb's comment assessment of polarity	0
subjectivity	double	Text Bolb's comment assessment of subjectivity	0
gspc	double	Was the stock or index mentioned? Will the user buy calls or puts? 0 for only mentioned, 1 if the user is buying calls or stock and -1 if they are buying puts or selling.	0
vix	double		
aapl	double		
dis	double		
tlsa	double		
nflx	double		
ba	double		
wmt	double		
amzn	double		
nvda	double		
movement	double	Does the user mention buying,selling or both? 1 for buying, -1 for selling and 0 for both.	0

Reference Slide - Subreddit Data Dictionary



Reference Slide - Data Collected

SVM True Vs Predicted - ^GSPC

Subreddit	Y	Pred Y: -1.5 or Less Delta %	-1.5 to -0.75 Delta %	-0.75 to 0.75 Delta %	0.75 to 1.5 Delta %	1.5 or Greater Delta %
No Subreddit Data	-1.5 or Less Delta %			9		
	-1.5 to -0.75 Delta %			36		
	-0.75 to 0.75 Delta %			492		
	0.75 to 1.5 Delta %			46	2	
	1.5 or Greater Delta %			4		1
StockMarket	-1.5 or Less Delta %	1		8		
	-1.5 to -0.75 Delta %		4	29		
	-0.75 to 0.75 Delta %			382		
	0.75 to 1.5 Delta %			38	8	
	1.5 or Greater Delta %			3		1

Reference - Classification Example

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